

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I, kadiri sankar Narayana (192325085) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

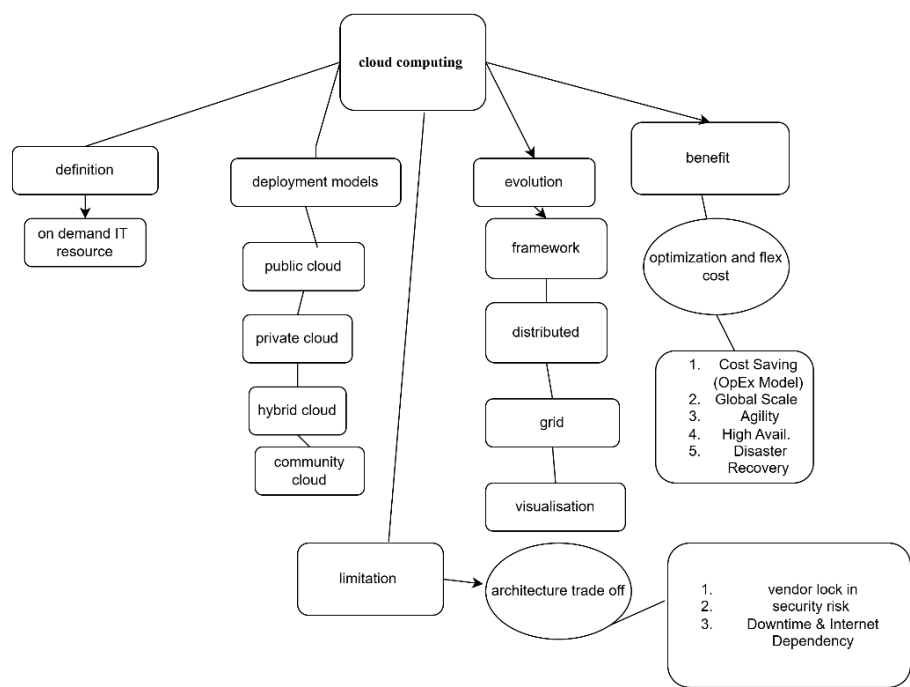
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

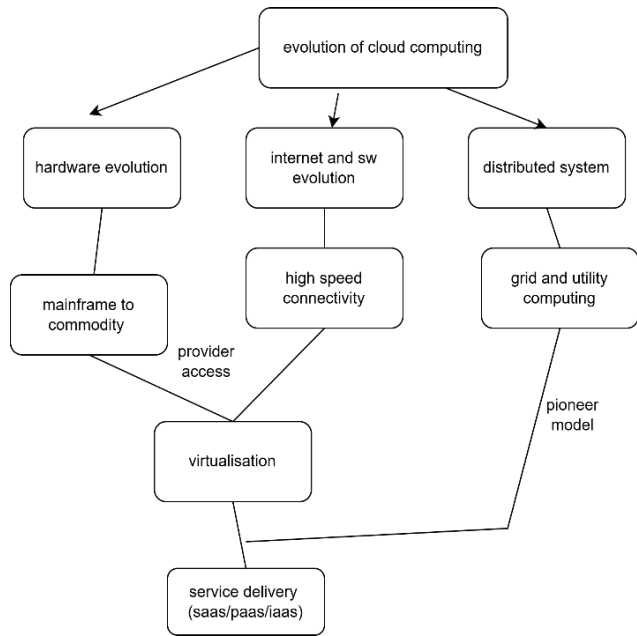
5. Questions

- i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
- ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
- iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
- iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
- v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

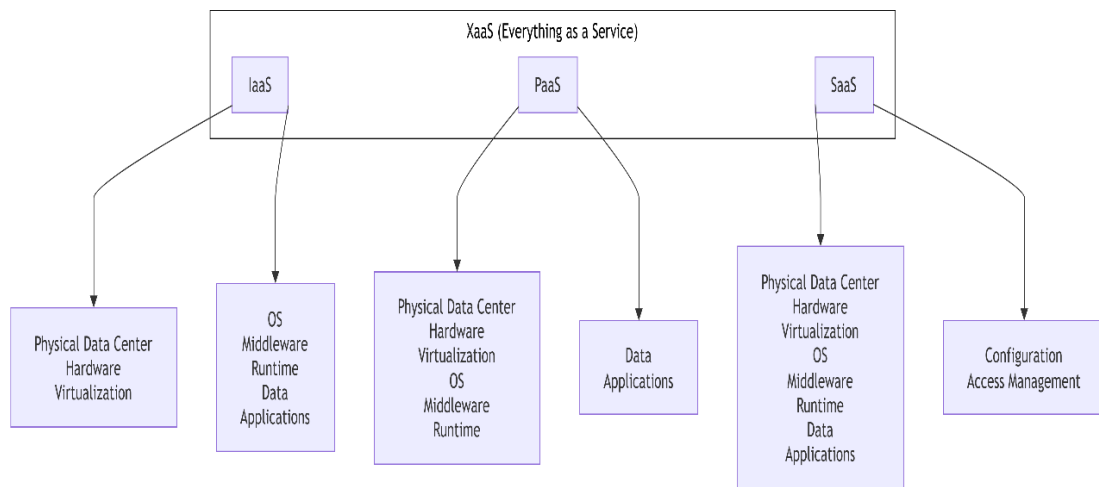
Concept map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



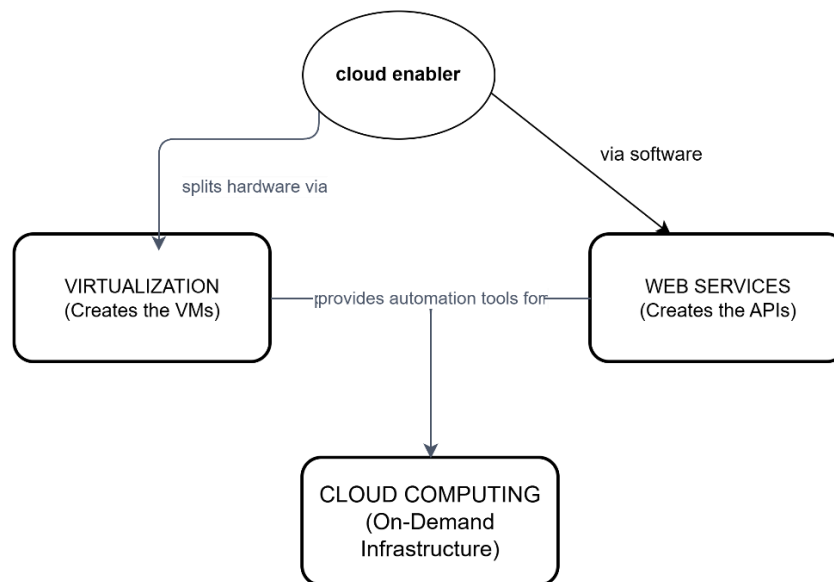
Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



5 questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The map works so well because it takes a sprawling, abstract concept like the cloud and anchors it into a clean, top-down hierarchy that matches how our brains naturally sort information. By cleanly boxing up big, overwhelming ideas into parallel columns, you can compare the pros and cons side-by-side without your eyes darting all over the screen. What really brings the whole thing together, though, are the small connecting words written right on the arrows. Instead of just guessing how the pieces fit together, those little words act like signposts, turning a basic collection of shapes into a clear, visual story that you can read straight through like a sentence.

ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, elasticity, and resource management serve as the structural backbone of the concept map, directly defining its Characteristics and enabling its Benefits.

- Resource management (via resource pooling) acts as the foundational layer, abstracting physical hardware into a shared pool to make multi-tenancy possible.
- Scalability dictates the architecture's capacity for long-term growth, ensuring it can scale out or up to handle heavier workloads.
- Elasticity provides the real-time automation loop, dynamically expanding or shrinking those resources on demand.

If these capabilities were absent, the architecture would collapse back into a rigid legacy system. Without resource management, the system faces the "noisy neighbor" syndrome, where one application hogs all bandwidth and starves the rest. Without scalability, the system hits a hard performance ceiling, leading to crashes under high traffic.

iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The inclusion of APIs, web services, and cloud models (IaaS/PaaS/SaaS) is justified because they provide the connectivity and abstraction layers required to transform rigid physical hardware into an automated, on-demand utility:

- APIs and Web Services act as the programmatic glue, allowing systems to communicate automatically and provision resources instantly without manual human intervention.
- Service Models (IaaS/PaaS/SaaS) cleanly divide operational responsibilities, abstracting infrastructure complexities based on user needs.

Together, they drive system efficiency by eliminating manual deployment overhead, enabling automated scaling, and drastically accelerating project deployment times.

iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The map's components interact through a direct cause-and-effect loop where core Characteristics actively drive the system's Benefits while balancing its Limitations. Specifically, the link between *Resource Pooling* and *Global Scalability* directly enhances performance, as data is distributed across dynamic clusters to keep latency low for users worldwide. Furthermore, the relationship between *Rapid Elasticity* and *Cost Saving* improves both user experience and financial efficiency by automating resource spikes during high traffic to prevent slowdowns, while scaling back down during drops to avoid wasting budget. Finally, the interaction between *Broad Network Access* and *Internet Dependency* exposes a critical reliability trade-off: it makes applications universally accessible, yet ties system availability entirely to the stability of the user's network connection.

v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Building this map really changed how I look at the cloud, moving me past the basic idea of "someone else's computer" to seeing it as a massive, beautifully synchronized ecosystem where business choices are directly tied to engineering rules. It clicked for me that things like saving money aren't just random perks; they are the direct results of technical features like resource pooling and auto-scaling working behind the scenes. If I were to redesign it, I'd break away from the neat, isolated columns and add a shared security boundary wrapped around the whole thing, while also visually layering the service models to show exactly where a provider's job ends and the user's responsibility begins.